DERIV	The DERIV subroutine firstly sets all previous concentrations (YP) to be Y from the previous time step. Emissions concentrations are updated according to dispersion effects captured in prior models. Then, the time derivative of species concentrations are determined through calculation of the production and loss of all 219 species. Flux rates for 130 key reactions are also calculated.	**YP(CELL,SPECIES)**: Previous total concentrations **EMI(CELL,EMI_SPECIES)**: Emissions concentrations **EMIP(CELL,EMI_SPECIES)**: Previous emissions concs. **RC(CELL,THERM_COEFFS)**: Thermal rate coefficients **DJ(CELL,PHOTOL_COEFFS)**: Photolysis rate coefficients **DTS**: Time step duration	**Y(CELL,SPECIES)**: Total concentrations **FL(CELL,REACTION)**: Flux rate of key reactions
WRITE	At the end of each time step, data is written back to the NetCDF file (boxm_ds.nc). This file then serves as input to the GPAT CHEM dataset.	**J(CELL,PHOTOL_PARAM)**: Photolysis parameters **DJ(CELL,PHOTOL_COEFF)**: Photolysis rate coefficients **Y(CELL,SPECIES)**: Total concentrations **FL(CELL,REACTION)**: Flux rate of key reactions	**CHEM**: Chemistry Dataset